THE LAWS OF SOFTWARE TEAMS

Price's law

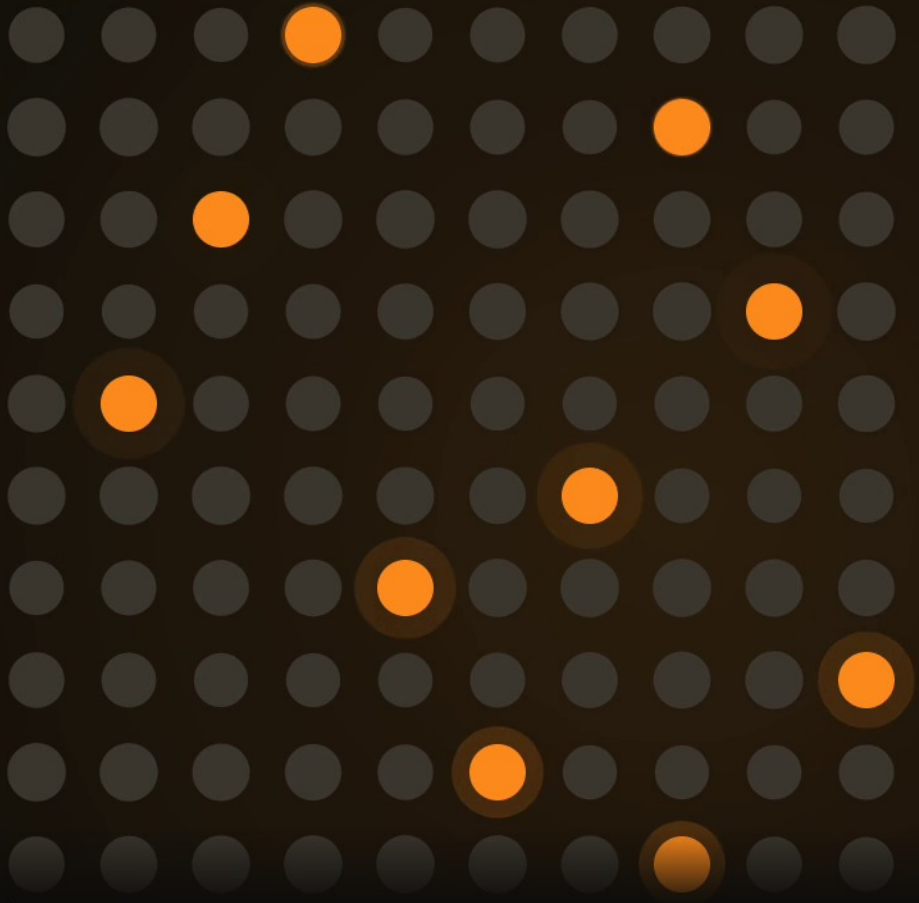
Half of the work comes from the square root of the people

swipe →

Ivan Rykov

01 THE HOOK

100 people on a team
10 of them do HALF the work



Price's Law

√N people
do half of the work



Derek J. de Solla Price

Physicist & historian of science, Yale

"Little Science, Big Science", 1963

The bigger the team, the smaller the core

team size → people doing half the work

10 people → 3 — 30%

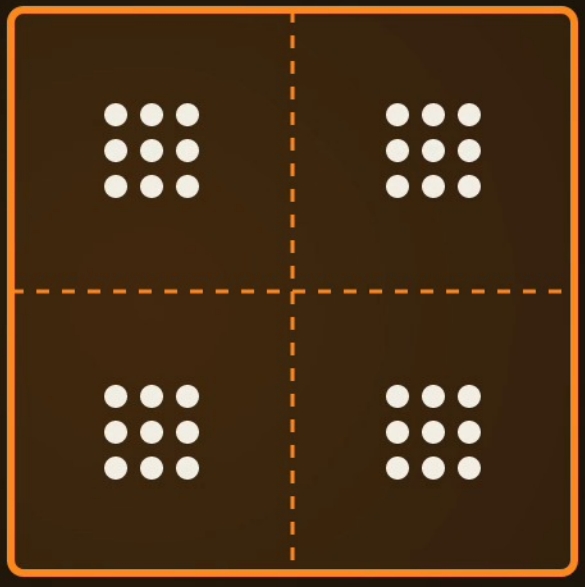
100 people → 10 — 10%

1,000 people → 32 — 3.2%

10,000 people → 100 — 1%

√N grows far slower than N

Want 2× the output? Hire 4× the people



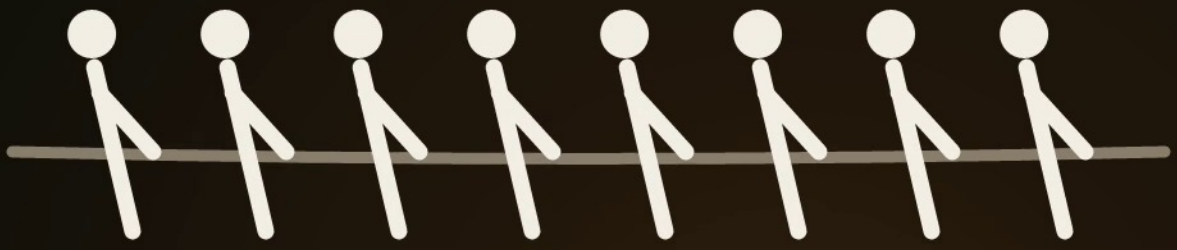
```
side = output  ×2
area = people  ×4
```

05 WHY

One reason:

the Ringelmann effect

1913 · pulling experiments



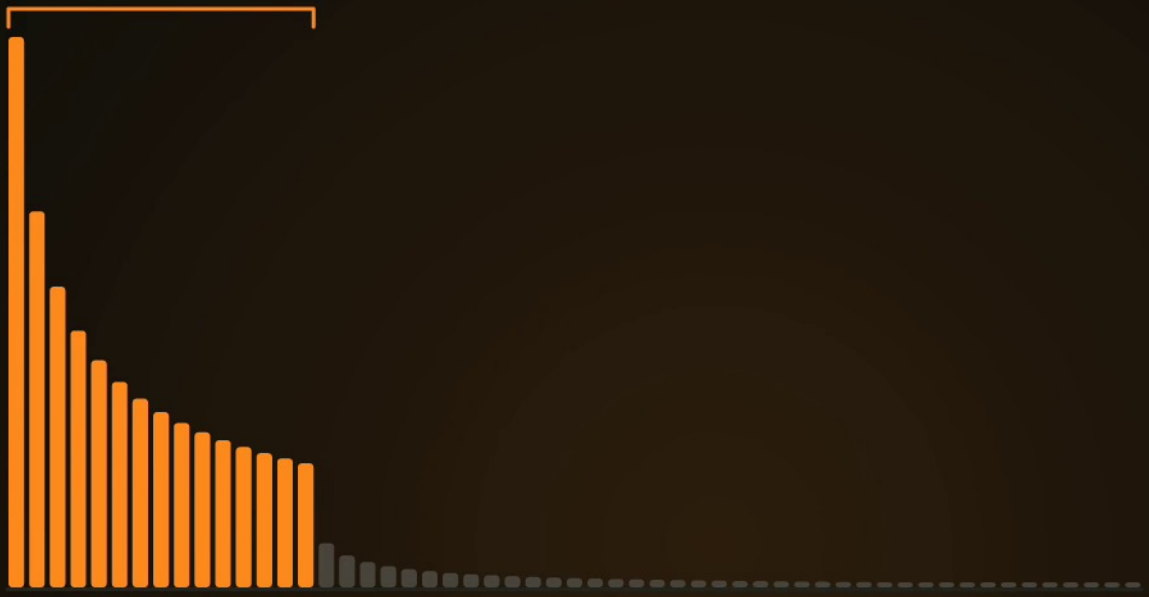


8 people 49% effort each

More people, less pull from each

In open source, it's even steeper

Apache web server · each bar = one developer



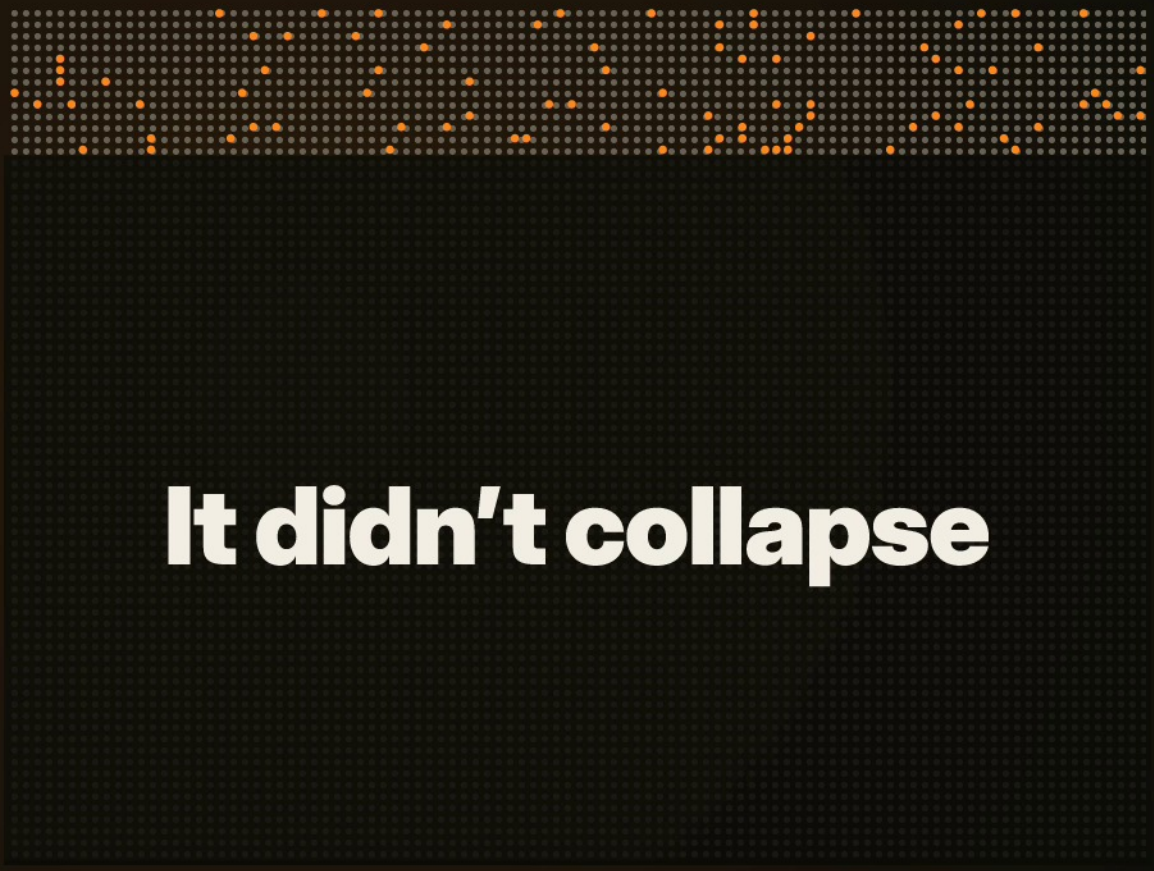
88% of new code came from the top 15

Mockus, Fielding & Herbsleb · Apache study, ICSE 2000

07 CASE STUDY

Twitter, late 2022

~7,500 employees



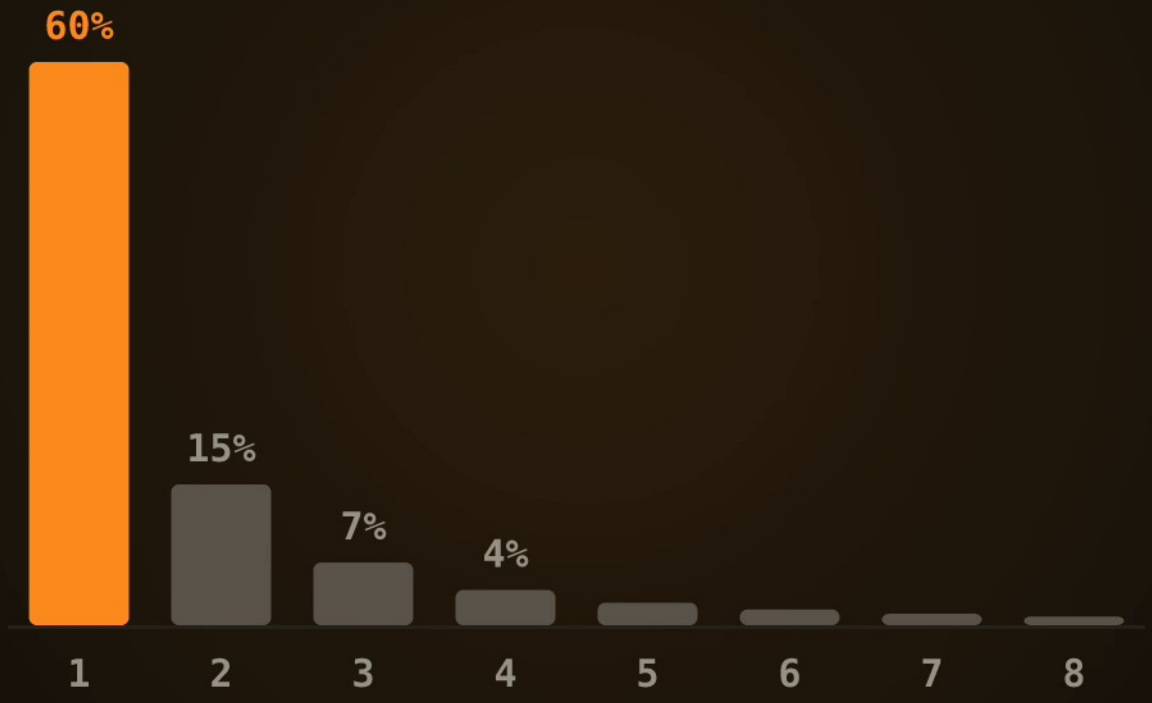
√7,500 ≈ 87 → half the work

Jan 2023: ~1,300 still working

Headcount: CNBC, Jan 2023

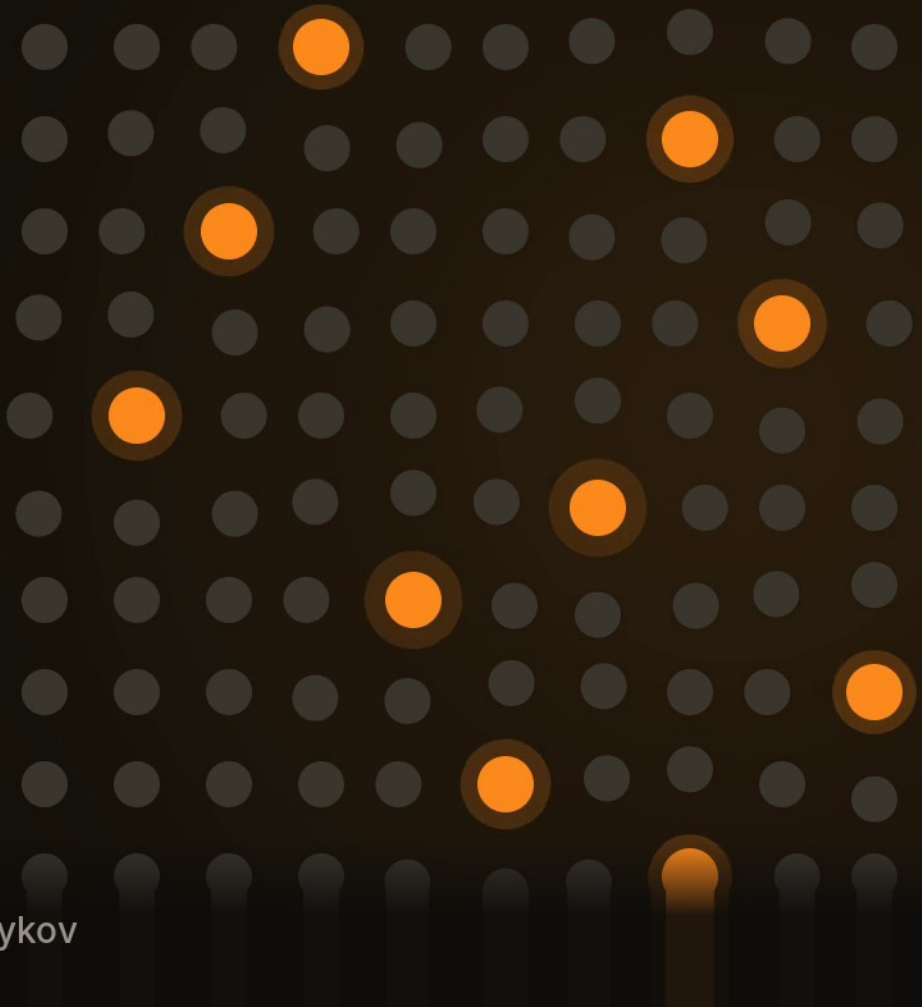
Plot twist:
it's not a real law

Real output is often even more lopsided



Lotka, 1926: 60% of authors publish only once

So don't count heads
Find your √N
Then clear their path



TAKEAWAY

So don’t count heads
Find your √N
Then clear their path

Sources: Derek de Solla Price, 1963 · Mockus, Fielding and Herbsleb, 2002 · Nicholls, 1988

next: Conway's law